

Am. 1o. que. no. 1

$$l = 0.25 \text{ m} , n = 36 \text{ cm}$$

$$(i) T = 2l = 0.5 \text{ s}$$

$$(ii) f = \frac{1}{T} = \frac{1}{0.5} \text{ Hz} = 2 \text{ Hz}$$

$$(iii) A = \frac{x}{2} = \frac{36}{2} = 18 \text{ cm}$$

Am. 1o. que. no. 2

$$m = 680 \text{ gm} = 0.68 \text{ kg}$$

$$k = 65 \text{ Nm}^{-1}$$

$$A = 11 \text{ cm} = 0.11 \text{ m}$$

$$l = 0$$

$$(i) \omega = \frac{2\pi}{T}$$

$$\Rightarrow T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}} \quad \left[\omega^2 = \frac{k}{m} \right]$$
$$= 2 \times 3.1416 \times \sqrt{\frac{0.68}{65}}$$

$$\therefore T = 0.643 \text{ s}$$

$$(ii) \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{65}{0.68}}$$

$$\therefore \omega = 9.777$$

$$\textcircled{iii} \quad x = 0.11 = A$$

$$x = A \cos(\omega t + \phi)$$

$$\therefore 0.11 = 0.11 \cos(\omega t + \phi)$$

$$\Rightarrow \cos(\omega t + \phi) = 1$$

$$\Rightarrow \cos(\omega t + \phi) = \cos(0)$$

$$\Rightarrow \cos(\omega \times 0 + \phi) = 0$$

$$\Rightarrow \phi = 0$$

$$\textcircled{iv} \quad t = 4 \text{ sec}, A = 0.11$$

$$v = -\omega A \sin(\omega t + \phi)$$

$$= -9.77 \times 0.11 \times \sin(9.77 \times 4 + 0) \text{ m s}^{-1}$$

$$= -0.6775 \text{ m s}^{-1}$$

$$v_{\max} = -\omega A = -9.77 \times 0.11 = 1.075 \text{ m s}^{-1}$$

$$\textcircled{v} \quad a_{(4 \text{ sec})} = -\omega^2 A \cos(\omega t)$$

$$= -(9.77)^2 \times 0.11 \times \cos(9.77 \times 4) \text{ m s}^{-2}$$

$$= -10.462 \text{ m s}^{-2}$$

$$a_{\max} = -\omega^2 A = -(9.77^2) \times 0.11$$

$$= -10.462 \approx -10.5$$

$$(vi) -t = 0,$$

$$x = A = 0.11$$

$$t = 7$$

$$\begin{aligned} x &= A \cos(\omega t) \\ &= 0.11 \cos(2.77 \times 7) \\ &= 0.0405 \text{ m} \end{aligned}$$

$$(vii) x = 0.11,$$

$$\begin{aligned} v &= \omega \sqrt{A^2 - x^2} \\ &= \omega \sqrt{(0.11)^2 - (0.11)^2} \\ &= 0 \text{ m s}^{-1} \end{aligned}$$

$$x = 0.04$$

$$\begin{aligned} v &= \omega \sqrt{A^2 - x^2} = 2.77 \sqrt{(0.11)^2 - (0.04)^2} \\ &= 1.00113 \text{ m s}^{-1} \end{aligned}$$

Ans. to que. no. 3

$$A = 8.5 \text{ cm} = 0.085 \text{ m}$$

$$m = 0.12 \text{ kg}$$

$$T = 0.2 \text{ s}$$

$$(i) f_{\max} = m a_{\max}$$

$$= m \omega^2 A$$

$$= m \left(\frac{2\pi}{T} \right)^2 A$$

$$= 0.12 \left(\frac{2 \times 3.14159}{0.2} \right)^2 \times 0.085 \text{ N}$$

$$= 10.067 \text{ N}$$

$$(i) F = -kx$$

$$\Rightarrow F_{\max} = |-kx_{\max}| = |-kA|$$

$$\Rightarrow k = \left| -\frac{F_{\max}}{A} \right| = \left| -\frac{10.067}{0.085} \right|$$

$$= 118.44$$

Am. to. que. no. 4

$$d = 0.002$$

$$f = 120 \text{ Hz}$$

$$\omega = 2\pi f = 240\pi = 753.98$$

$$(i) A = x_{\max} = \frac{d}{2} = \frac{0.002}{2} = 0.001 \text{ m}$$

$$(ii) v_{\max} = -\omega A = -753.98 \times 0.001 \text{ m}^{-1}$$

$$= -0.75398 \text{ m}^{-1}$$

$$= -0.754 \text{ m}^{-1}$$

$$(iii) a_{\max} = -\omega^2 A = -(753.98)^2 \times 0.001 \text{ m}^{-2}$$

$$= -568.486 \text{ m}^{-2}$$

Am. to. que. no. 5

$$k = 400, x = 0.1 \text{ m}, v = -13.6 \text{ m}^{-1}, a = -123 \text{ m}^{-2}$$

$$(i) f = \frac{\omega}{2\pi} = \sqrt{\frac{k}{m}} \times \frac{1}{2\pi} = \sqrt{\frac{k}{-(kx)/a}} \times \frac{1}{2\pi}$$

$$= \sqrt{-\frac{a}{x}} \times \frac{1}{2\pi} = \sqrt{\frac{-(-123)}{0.1}} \times \frac{1}{2 \times 3.14159}$$

$$= 5.582 \text{ Hz}$$

$$F = ma = -kx$$

$$\Rightarrow m = -\frac{kx}{a}$$

$$\begin{aligned}
 \textcircled{ii} \quad m &= -\frac{kx}{a} = -\frac{400 \times (-12.3)}{0.1} \text{ kg} \\
 &= \frac{-400 \times (0.1)}{(-12.3)} \text{ kg} \\
 &= 0.325 \text{ kg}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{iii} \quad v &= \omega \sqrt{A^2 - x^2} \\
 \Rightarrow v^2 &= \omega^2 (A^2 - x^2) \\
 \Rightarrow A^2 &= \frac{v^2}{\omega^2} + x^2 \\
 \Rightarrow A &= \sqrt{\frac{v^2}{\omega^2} + x^2} = \sqrt{\frac{(-13.6)^2}{(2\pi(5.582))^2} + (0.1)^2} \text{ m} \\
 &= 0.4004 \text{ m}
 \end{aligned}$$

Am. to que no - 6

$$m = \cancel{0.25} \text{ kg} \quad 25 \text{ g} = 0.025 \text{ kg}$$

$$k = 400 \text{ dynes/cm}$$

$$= \cancel{0.004} \quad 0.4 \text{ Nm}^{-1} \quad [1 \text{ dyne/cm} = 0.001 \text{ N/m}]$$

$$x = 10 \text{ cm} = 0.1 \text{ m}$$

$$\begin{aligned}
 \textcircled{i} \quad T &= \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}} = 2 \times 3.14159 \sqrt{\frac{0.025}{0.4}} \\
 &= 1.571 \text{ s}
 \end{aligned}$$

$$\textcircled{ii} \quad f = \frac{1}{T} = 0.637 \text{ Hz}$$

$$\textcircled{iii} \quad \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{0.4}{0.025}} = 4$$

$$\textcircled{iv} \quad v_{\max} = \omega A = 4 \times 0.1 = 0.4 \text{ ms}^{-1}$$

Am. to. que no - 7

$$m = 1.68 \times 10^{-27} \text{ kg}$$

$$f = 10^{14} \text{ Hz}$$

$$A = 10^{-10} \text{ m}$$

$$f_{\text{max}} = m a_{\text{max}} = m \omega^2 A = m (2\pi f)^2 A$$

$$= 1.68 \times 10^{-27} \times 4 \times (3.14159)^2 \times (10^{14})^2 \times 10^{-10} \text{ N}$$

$$= 6.632 \times 10^{-8} \text{ N}$$

Am. to. que. no - 8

$$a_{\text{max}} = -\omega^2 A \quad \text{--- (i)}$$

$$v_{\text{max}} = -\omega A \quad \text{--- (ii)}$$

$$\text{(i)} \div \text{(ii)} \rightarrow$$

$$\frac{a_{\text{max}}}{v_{\text{max}}} = \frac{-\omega^2 A}{-\omega A}$$

$$\Rightarrow \omega = \frac{a_{\text{max}}}{v_{\text{max}}} = \frac{1.57}{1} = 1.57$$

$$T = \frac{2\pi}{\omega} = \frac{2 \times 3.14159}{1.57} = 4.002 \text{ s}$$

Am. - lo. que. no. 9

$$A = 5 \text{ m}$$

$$x = 3 \text{ m}$$

$$a = 48 \text{ m}^{-2} \text{ s}^{-2} \text{ --- } \omega^2 A \cos$$

$$\textcircled{1} \quad \cancel{v = A^2} \quad a = -\omega^2 \sqrt{A^2 - x^2}$$

$$\Rightarrow 48 = \omega^2 \sqrt{25 - 9}$$

$$\Rightarrow \omega^2 = \frac{48}{4} = 12$$

$$\Rightarrow \omega = 3.464$$

$$\therefore v = \omega \sqrt{A^2 - x^2}$$

$$= 3.464 \times 4$$

$$= 13.856 \text{ m}^{-1}$$

$$\textcircled{u} \quad T = \frac{2\pi}{\omega} = \frac{2 \times 3.14159}{3.464}$$

$$= 1.814 \text{ s}$$

$$\textcircled{u} \quad v_{\max} = \omega A$$

$$= 3.464 \times 5$$

$$= 17.32 \text{ m}^{-1}$$

Am. 10. que. no- 10

$$x_{\max} = A = 0.37 \text{ cm} = 3.7 \times 10^{-3} \text{ m}$$

$$f = 0.25 \text{ Hz}$$

$$\textcircled{i} \quad T = \frac{1}{f} = \frac{1}{0.25} = 4 \text{ s}$$

$$\textcircled{ii} \quad \omega = 2\pi f = 2 \times 3.14159 \times 0.25 \\ = 1.571 \text{ rad/sec}$$

$$\textcircled{iii} \quad A = x_{\max} = 0.0037 \text{ m} = 3.7 \times 10^{-3} \text{ m}$$

$$\textcircled{iv} \quad t = 3.0 \text{ s}$$

$$x = A \cos(\omega t) \\ = 3.7 \times 10^{-3} \times \cos(1.571 \times 3) \\ = 3.687 \times 10^{-3} \text{ m}$$

$$\textcircled{v} \quad t = 3.0 \text{ s}$$

$$v = -\omega A \sin(\omega t) = -1.571 \times 0.0037 \times \sin(1.571 \times 3) \text{ m} \\ = -4.776 \times 10^{-4} \text{ m/s}$$

Am. 10. que. no- 11

$$x = 10 \cos\left(3\pi t + \frac{\pi}{3}\right) \text{ m}, \quad A = 10 \text{ m}, \quad \omega = 3\pi$$

$$\textcircled{i} \quad t = 2.5 \text{ s}$$

$$x = 10 \cos(3\pi \times 2.5 + 60) \text{ m} \\ = 1.1213 \text{ m}$$

(i) $t = 3.0 \text{ s}$

$$v = -\omega A \sin(\omega t + \phi)$$

$$= -3\pi \times 10 \sin\left(3\pi \times 3 + \frac{\pi}{3}\right)$$

$$= -94.205 \text{ m s}^{-1}$$

(ii) $t = 2.0 \text{ s}$

$$a = -\omega^2 A \cos(\omega t + \phi)$$

$$= -(3\pi)^2 \times 10 \cos\left(3\pi \times 2 + \frac{\pi}{3}\right)$$

$$= -90\pi^2 \cos\left(6\pi + \frac{\pi}{3}\right)$$

$$= -171.78 \text{ m s}^{-2}$$

(iv) $v_{\max} = -\omega A$

$$= -3\pi \times 10$$

$$= -94.25 \text{ m s}^{-1}$$

Ans. to que. no-12

$$x = 10 \sin\left(10t - \frac{\pi}{6}\right); A = 10; \omega = 10; \phi = -\frac{\pi}{6}$$

(i) $f = 2\pi \quad f = \frac{\omega}{2\pi} = \frac{10}{2\pi} = 1.59 \text{ Hz}$

(ii) $T = \frac{1}{f} = 0.629 \text{ s}$

(iii) $x_{\max} = A = 10 \text{ m}$

(iv) $v_{\max} = \omega A = 10 \times 10 \text{ m s}^{-1} = 100 \text{ m s}^{-1}$

(v) $a_{\max} = \omega^2 A = (10)^2 \times 10 \text{ m s}^{-2} = 1000 \text{ m s}^{-2}$

Ans. to que. no - 13

$$A = 4 \text{ m}$$

$$\omega = 2\pi f = 2 \times \pi \times 0.5 = \pi$$

$$f = 0.5 \text{ Hz}$$

$$\phi = \frac{\pi}{4} = 45^\circ$$

$$(i) T = \frac{1}{f} = \frac{1}{0.5} = 2 \text{ s}$$

$$(ii) x = A \cos(\omega t + \phi)$$

$$\therefore x = 4 \cos\left(\pi t + \frac{\pi}{4}\right)$$

$$(iii) t = 5 \text{ s}$$

$$v = -\omega A \sin(\omega t + \phi)$$

$$= -4\pi \sin\left(\pi t + \frac{\pi}{4}\right)$$

$$= -2.857 \text{ m s}^{-1}$$

$$a = -\omega^2 A \cos(\omega t + \phi)$$

$$= -4\pi^2 \cos\left(5\pi + \frac{\pi}{4}\right)$$

$$= -19.32 \text{ m s}^{-2}$$

Ans. to que. no - 14

$$m = 2 \text{ kg} ; k = 196 \text{ N/m} ; x_{\text{max}} = 5 \text{ cm} = 0.05 \text{ m}$$

$$(i) \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{196}{2}} = 9.89 \text{ rad/sec}$$

$$f = \frac{\omega}{2\pi} = \frac{9.89}{2 \times 3.14159} = 1.57 \text{ Hz}$$

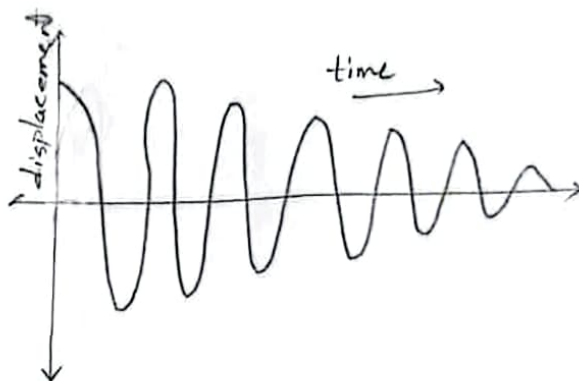
$$T = \frac{1}{f} = \frac{1}{1.57} = 0.637 \text{ s}$$

ii) $x = A \cos(\omega t)$ $[A = x_{\max} = 0.05]$

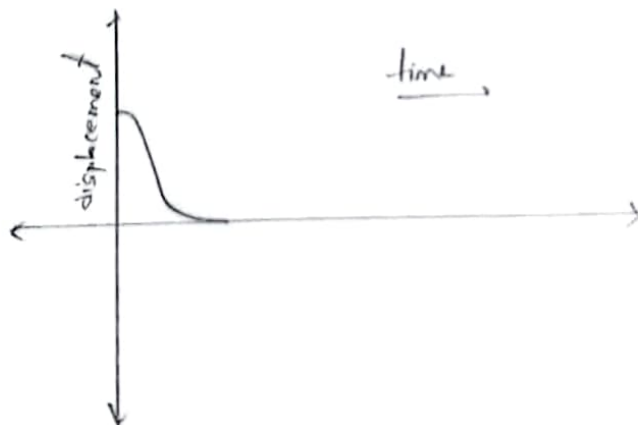
$\therefore x = 0.05 \cos(9.89t)$

Am. to. que. no - 16

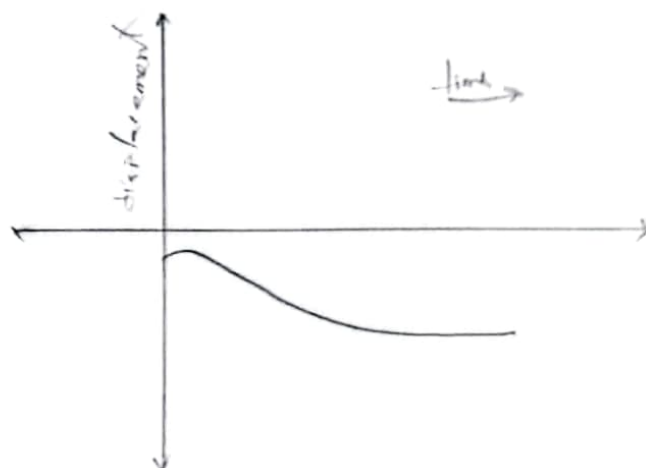
$\omega/\gamma = 10$ (Oscillatory)



$\omega/\gamma = 0.5$ (Critical)



$\omega/\gamma = 0.03$



Am. to. que. no-15

$$m = 2.72 \times 10^5 \text{ kg} ; f = 10 \text{ Hz} ; A = x_m = 20 \text{ cm} = 0.2 \text{ m}$$

$$\textcircled{i} E = \frac{1}{2} k A^2$$

$$= \frac{1}{2} \times \omega^2 m A^2 = \frac{1}{2} (2\pi f)^2 m A^2 = \frac{1}{2} 4\pi^2 f^2 m A^2$$

$$= 2\pi^2 f^2 m A^2$$

$$= 2 \times (3.14159)^2 \times 10^2 \times 2.72 \times 10^5 \times (0.2)^2$$

$$\textcircled{ii} F = \frac{1}{2} k x_m^2$$

$$= \frac{1}{2} \times 1.07 \times 10^9 \times (0.2)^2$$

$$= 2.14 \times 10^7$$

$$\omega = 2\pi f = 62.83$$

$$k = \omega^2 m = 1.07 \times 10^9$$

$$\textcircled{iii} v_{\max} = \omega A$$

$$= 62.83 \times 0.2$$

$$= 12.57 \text{ m s}^{-1}$$

Am. to. que. no-17

$$L = 0.4 \text{ H} , C = 0.0020 \mu\text{F} = 2 \times 10^{-9} \text{ F}$$

$$\textcircled{i} R_{\max} = 2 \sqrt{\frac{L}{C}} = 2 \sqrt{\frac{0.4}{2 \times 10^{-9}}} = 28284.27$$

$$\textcircled{ii} f = \frac{1}{2\pi} \times \frac{1}{\sqrt{LC}} = 5626.982$$



Am. to. que. no - 18

$$C = 0.1 \mu F = 10^{-7} F ; L = 10 \text{ mH} = 0.01 \text{ H} ; R = 200 \Omega$$

$$\frac{1}{LC} = \frac{1}{0.01 \times 10^{-7}} = 10^9$$

$$\frac{R^2}{4L^2} = \frac{(200)^2}{4 \times (0.01)^2} = 10^8$$

$$\therefore \frac{1}{LC} > \frac{R^2}{4L^2} \therefore \text{Oscillatory}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{1}{LC}} = \frac{1}{2 \times 3.14156} \sqrt{10^9} = 5032.725 \text{ Hz}$$

Am. to. que. no - 19

$$m = 250 \text{ gm} = 0.25 \text{ kg} ; k = 85 \text{ N/m} ; b = 70 \text{ gm/s} = 0.07 \text{ kg s}^{-1}$$

$$\omega = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} = 18.44$$

$$(i) T = \frac{2\pi}{\omega} = 0.341$$

$$(ii) t_{1/2A} = \frac{2m}{b} \ln 2 = 4.95$$

$$(iii) N = \frac{t_{1/2}}{T} = 14.5 \approx 15$$

$$(iv) \gamma = \frac{m}{b} = \frac{0.25}{0.07} = 3.5714$$

$$(v) x_{max} = 0.35$$

$$E_r = \frac{1}{2} k A^2 (e^{-bt/m}) = 5.187$$

$$(vi) A = A_0 e^{-\lambda t} \therefore \frac{A}{A_0} = e^{-\frac{bt}{m}}$$

Ans. to. que. no. 20

$$y = 4.0 \sin(0.10x - 2t)$$

(i) $A = 4$

(ii) $\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.1} = 20\pi$

(iii) $v = \frac{\omega}{k} = \frac{-2}{-0.1} = 20$

(iv) $\phi = \frac{\omega}{2\pi} = -\frac{2}{2\pi} = -\frac{1}{\pi} = -0.318$

Ans. to. que. no. 21

$$y = 4.0 \sin 2\pi \left(\frac{x}{100} \right)$$

$$v = 30 \text{ cm/s} = 0.3 \text{ m/s}$$

$$T = 3 \text{ s}$$

$$f = \frac{v}{\lambda} = 0.3$$

$$\omega = 2\pi f = 2\pi \times 0.3 = 0.6\pi = 1.885$$

$$y = 4.0 \sin \left(\frac{2\pi x}{100} - 0.6\pi \times 0.3 + 2\pi \right)$$

$$\therefore y = 4.0 \sin \left(\frac{2\pi x}{100} + \frac{\pi}{5} \right)$$

Am. to. que. no. 22

$$y = 5.0 \sin \pi (360 t - 0.15 x) \quad \left| \quad \frac{2}{\lambda} = 0.15 \therefore \lambda = 13.33 \right.$$
$$= 5 \sin (360 \pi t - 0.15 \pi x)$$

(i) $A = 5$

(ii) $y = 5 \sin \frac{2\pi}{\lambda} (180 t - 0.075 x)$

$$0.075 = \frac{1}{\lambda}$$

$$\Rightarrow \lambda = \frac{1}{0.075} = 13.33$$

$$180 = \frac{v}{\lambda}$$

$$\Rightarrow v = 180 \times \lambda = 180 \times 13.33 = 240$$

(iii) $\lambda = 13.33$

(iv) $f = \frac{v}{\lambda} = \frac{240}{13.33} = 180$

(v) $T = \frac{1}{f} = \frac{1}{180} = 5.55 \times 10^{-3}$

Am. to. que. no. 23

$$y = A \sin \frac{2\pi}{\lambda} (vt - x)$$

$$\Rightarrow \frac{y}{A} = \sin 2\pi \left(\frac{vt}{\lambda} - \frac{x}{\lambda} \right) = \sin 2\pi \left(ft - \frac{x}{\lambda} \right)$$

$$\Rightarrow \frac{y}{A} = \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) \dots (1)$$

Am. to. que. no. 22

$$y = 5.0 \sin \pi (360 t - 0.15 x) \quad \left| \quad \frac{2}{\lambda} = 0.15 \therefore \lambda = 13.33 \right.$$
$$= 5 \sin (360 \pi t - 0.15 \pi x)$$

(i) $A = 5$

(ii) $y = 5 \sin \frac{2\pi}{\lambda} (180 t - 0.075 x)$

$$0.075 = \frac{1}{\lambda}$$

$$\Rightarrow \lambda = \frac{1}{0.075} = 13.33$$

$$180 = \frac{v}{\lambda}$$

$$\Rightarrow v = 180 \times \lambda = 180 \times 13.33 = 240$$

(iii) $\lambda = 13.33$

(iv) $f = \frac{v}{\lambda} = \frac{240}{13.33} = 180$

(v) $T = \frac{1}{f} = \frac{1}{180} = 5.55 \times 10^{-3}$

Am. to. que. no. 23

$$y = A \sin \frac{2\pi}{\lambda} (vt - x)$$

$$\Rightarrow \frac{y}{A} = \sin 2\pi \left(\frac{vt}{\lambda} - \frac{x}{\lambda} \right) = \sin 2\pi \left(ft - \frac{x}{\lambda} \right)$$

$$\Rightarrow \frac{y}{A} = \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) \dots (1)$$

$$y_1 = 6, A = 8, x_1 = 10 \text{ cm}$$

$$\frac{6}{8} = \sin 2\pi \left(\frac{t}{T} - \frac{0.10}{\lambda} \right)$$

$$\Rightarrow 2\pi \left(\frac{t}{T} - \frac{0.10}{\lambda} \right) = \sin^{-1} \left(\frac{6}{8} \right) = 48.59$$

$$\Rightarrow \frac{t}{T} - \frac{0.1}{\lambda} = 7.733 \dots \text{--- (i)}$$

$$y_2 = 4, A = 8, x_2 = 25$$

$$\frac{4}{8} = \sin 2\pi \left(\frac{t}{T} - \frac{0.25}{\lambda} \right)$$

$$\Rightarrow 2\pi \left(\frac{t}{T} - \frac{0.25}{\lambda} \right) = \sin^{-1} \left(\frac{1}{2} \right) = 30$$

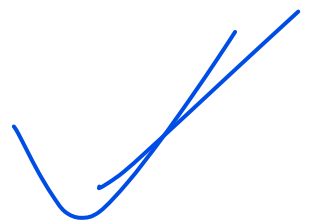
$$\Rightarrow \frac{t}{T} - \frac{0.25}{\lambda} = 4.775 \text{ --- (ii)}$$

$$\text{(i)} - \text{(ii)} \rightarrow$$

$$\frac{t}{T} - \frac{0.1}{\lambda} - \frac{t}{T} + \frac{0.25}{\lambda} = 7.733 - 4.775$$

$$\Rightarrow \frac{0.15}{\lambda} = 2.958$$

$$\Rightarrow \lambda = \frac{0.15}{2.958} = 0.051 \text{ m} = \boxed{5.1 \text{ cm}}$$



-Am. 10. que. no. 24

$$T = 5.00 \text{ s}$$

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$\Rightarrow L = \frac{T^2 g}{4\pi^2} = \boxed{6.21 \text{ m}}$$

$$f = \frac{1}{T} = \frac{1}{5} = 0.2 \text{ Hz}$$

-Am. 10. que. no. 25

$$y_1 = A \cos\left(\frac{1}{3} kx + \omega t\right) \dots \textcircled{I}$$

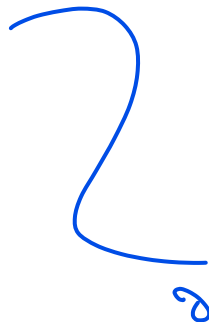
$$y_2 = A \cos\left(\frac{1}{3} kx - \omega t\right) \dots \textcircled{II}$$

$$\textcircled{I} + \textcircled{II} \rightarrow$$

$$y_1 + y_2 = A \cos\left(\frac{1}{3} kx + \omega t\right) + A \cos\left(\frac{1}{3} kx - \omega t\right)$$

$$= A \left(\cos\left(\frac{1}{3} kx + \omega t\right) + \cos\left(\frac{1}{3} kx - \omega t\right) \right)$$

$$= \boxed{2A \cos\left(\frac{1}{3} kx\right)} \cos(\omega t)$$



Am. to. que. no - 26

$$x = Ae^{-\alpha t} \cos(\omega_d t + \delta)$$

$$v = -\omega_d Ae^{-\alpha t} \sin(\omega_d t + \delta)$$

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

$$= \frac{1}{2}m \left[\omega_d^2 A^2 e^{-2\alpha t} \sin^2(\omega_d t + \delta) \right] + \frac{1}{2}k A^2 e^{-2\alpha t} \cos^2(\omega_d t + \delta)$$

$$= \frac{1}{2}k A^2 e^{-2\alpha t} \left[\sin^2(\omega_d t + \delta) + \cos^2(\omega_d t + \delta) \right] \quad \left[\omega^2 = \frac{k}{m} \Rightarrow k = m\omega^2 \right]$$

$$= \frac{1}{2}k A^2 e^{-2\alpha t}$$

$$= E_0 e^{-2\alpha t}$$

$$\therefore E = E_0 e^{-2\alpha t}$$

[allowed]

Am. to. que. no - 27

$$C_1 = 2 \text{ cm/s} = 0.02 \text{ m/s}$$

$$\lambda_1 = 4 \text{ cm} = 0.04 \text{ m}$$

$$C_2 = 3 \text{ cm/s} = 0.03 \text{ m/s}$$

$$\lambda_2 = 3 \text{ cm} = 0.03 \text{ m}$$

$$C_1 = \frac{\omega_1}{k} = \frac{\omega_1 \lambda_1}{2\pi} \Rightarrow \boxed{\omega_1 = 3.1416 \text{ rad/s}}$$

$$C_2 = \frac{\omega_2 \lambda_2}{2\pi} \Rightarrow \boxed{\omega_2 = 6.28 \text{ rad/s}}$$

$$v_1 = 0 \text{ m/s}$$

$$v_2 = 0 \text{ m/s}$$

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Am. 10. que. no. 28

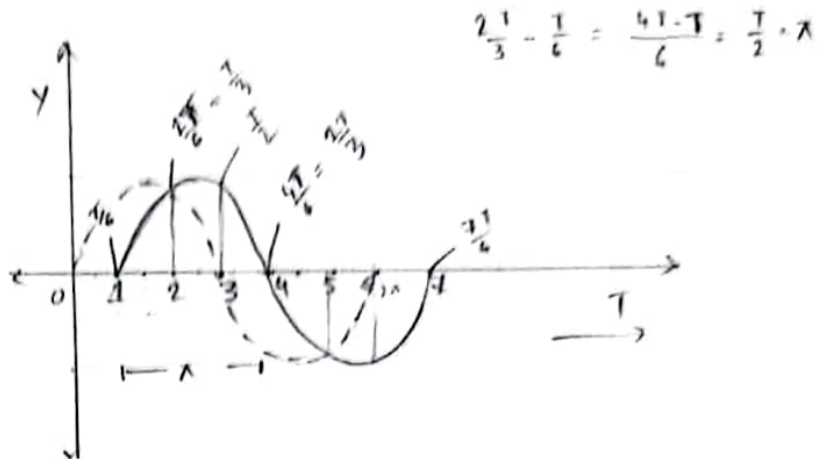
i) $y = A \sin(\omega t - \frac{\pi}{3})$

$\phi = -\frac{\pi}{3}$

$T = 2\pi \Rightarrow \frac{T}{2} = \pi$

$\Rightarrow \frac{T}{6} = \frac{\pi}{3}$

$\Rightarrow \frac{T}{6} \times 3 = \pi$



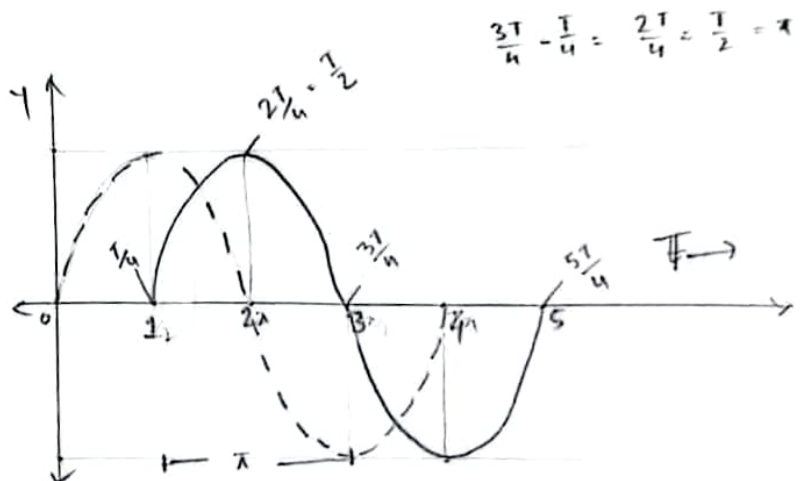
ii) $y = A \sin(\omega t - \frac{\pi}{2})$

$\phi = -\frac{\pi}{2}$

$T = 2\pi \Rightarrow \frac{T}{2} = \pi$

$\Rightarrow \frac{T}{4} = \frac{\pi}{2}$

$\Rightarrow \frac{T}{4} \times 2 = \pi$

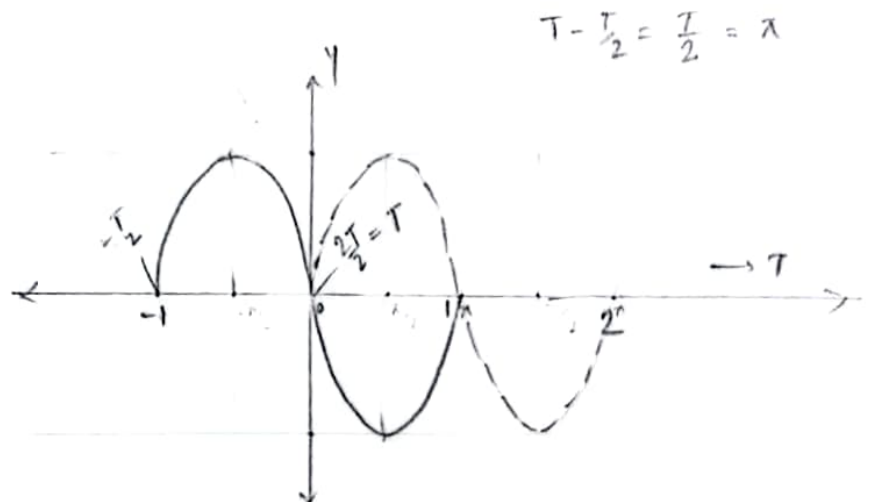


iii) $y = A \sin(\omega t + \pi)$

$\phi = \pi$

$\frac{T}{2} = \pi$

$\Rightarrow -\frac{T}{2} = -\pi$

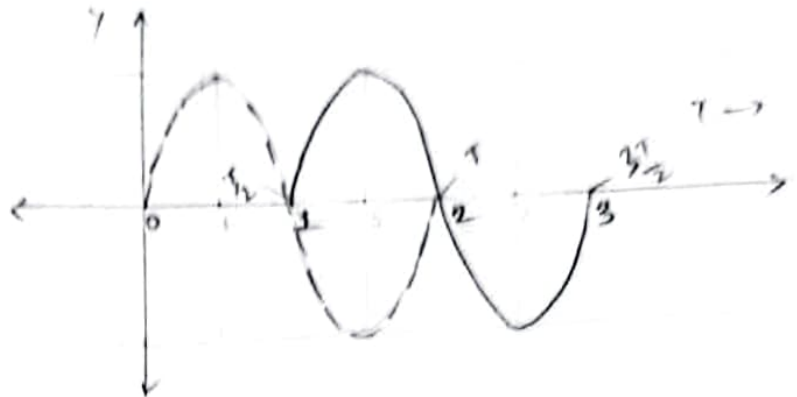


(iv) $y = A \sin(\omega t - \pi)$

$$T - \frac{T}{2} = \frac{T}{2} = \pi$$

$$\phi = -\pi$$

$$\frac{T}{2} = \pi$$



(v) $y = A \sin(\omega t + \frac{3\pi}{4})$

$$\phi = \frac{3\pi}{4}$$

$$T = 2\pi$$

$$\Rightarrow \frac{T}{2} = \pi$$

$$\Rightarrow -\frac{3T}{8} = -\frac{3\pi}{4}$$

$$\begin{aligned} -\frac{3T}{8} + \frac{T}{2} &= \frac{-3T + 4T}{8} \\ &= \frac{T}{8} \end{aligned}$$

$$\frac{T}{8} - \left(-\frac{3T}{4}\right) = \frac{T}{8} + \frac{3T}{8} = \frac{4T}{8} = \frac{T}{2} = \pi$$

